MAPPA: MONITORING AND ANALYSIS OF PLASTIC POLLUTION IN AQUATIC ENVIRONMENTS

Sampling of microplastics in water: LOTIC environments

Ver. 1

1. Objectives:

a) Training and applying a standardized protocol for the identification and quantification of microplastics in freshwater ecosystems

b) Analysis of the relationship between the presence of microplastics and the climatic variables, geomorphological characteristics, land use and anthropogenic impact.

c) Contribute the data obtained to the public, with the aim of facilitating the formulation of policies for plastic pollution mitigation.

2. Schedule

Phase 1 (2023-2024): From December 2023 to March 2024, the first coordinated sampling will be conducted, with one sampling per study site.

Phase 2 (2024-2025): The sampling frequency will be increased to address specific questions derived from the results obtained in the stage 1.

3. Participation and publications

a) Based on the data collected (including previous data), publications will be carried out in scientific journals. Those who participate in the sampling of one site will be entitled to two **co-authorships**, with an additional (+1 per extra site) for those who contribute to two or more sites. However, we recognize the possible labor difficulties, so this criterion is flexible and open to discussion when necessary.

b) b) The **order of merit** will be established alphabetically, with consideration of those who participate in additional activities (analyzing additional samples, data analysis, content creation, etc.) In such circumstances, these additional contributions will be considered in determining the order of authorship.

4. Sampling protocol options

A. Sites with higher flow (> 0.5 m depth): plankton net (page 3)
B. Sites with low flow (< 0.5 m depth): bucket and plankton net (page 6)

4.1. Materials:

- Plankton net (pore size < 50 µm) with a cotton bag.
- Flowmeter (recommended).
- Metal bucket (protocol B).
- Three plastic (PVC recommended) or glass containers bottle per study site (sample, replicate, blank/negative).
- GPS (Wikiloc app).
- 5 liters of Milli Q or distilled water per site (Blank sample).
- water to concentrate the sample and clean the net (tap water or from the same waterbody).
- Fridge (up to 2-3 months storage).

4.2. To be considered:

- **Weather/extreme events:** It is recommended to carry out the sampling on calm days, preferably in the early morning, with low wind speed (<20 km/h) and no precipitation. In case of storms, wait at least two days before sampling. In the case of extreme weather events, sampling will be postponed until the situation normalizes.
- **Location:** A sampling point close to the center of the waterbody or in the deepest possible location will be selected. In the presence of urban centers (if any), priority will be given to points close to them.
- **Turbidity:** In situations of high concentration of organic matter, the sampling will be repeated until the desired volume is obtained, preventing the net from clogging.
- **Please, take some pictures to share with the MappA community!**

4.3. Sampling

BLANK/NEGATIVE: 5 liters of Milli Q or distilled water will be filtered using the same sampling net (before the survey). Then, the sample will be collected in a plastic or glass container and labeled as "**Blank_1 ddmmyyyy**", indicating the corresponding date (dd/mm/yyyy)

A- Sites with higher flow (> 0.5 m depth): PLANKTON NET

Use a pre-cleaned plankton net for sampling. Transport it in a cotton bag to avoid contamination by microplastics, especially fibers.

1. Collect a sample of surface water in the highest flow area while maintaining a plankton net at a fixed point. Only 3/4 of the net will be submerged (Fig. 1), and it is recommended to mark the mouth of the net to indicate the part that should remain above the water, considering the heights (H) mentioned in Table 1.
2. Record the coordinates of the sampling point using GPS. If you do not have a GPS device, it is suggested to use the "Wikiloc" application on a cell phone or tablet to record the coordinates.
3. Whenever possible, use a flowmeter to calculate the water volume. This will be secured with ropes below 3/4 of the net, in a central position (Fig. 1). Filter a minimum of 20 m3, according to the necessary time indicated in Table 2 based on the diameter of the net and the current velocity. If the net becomes clogged before reaching this volume, conduct multiple samplings until the desired volume is achieved. Record the time of each collection to calculate the filtered partial volume and consolidate all samples in the same container. It is important to note that 3/4 of the net must remain submerged for precise sampling (Fig. 1).

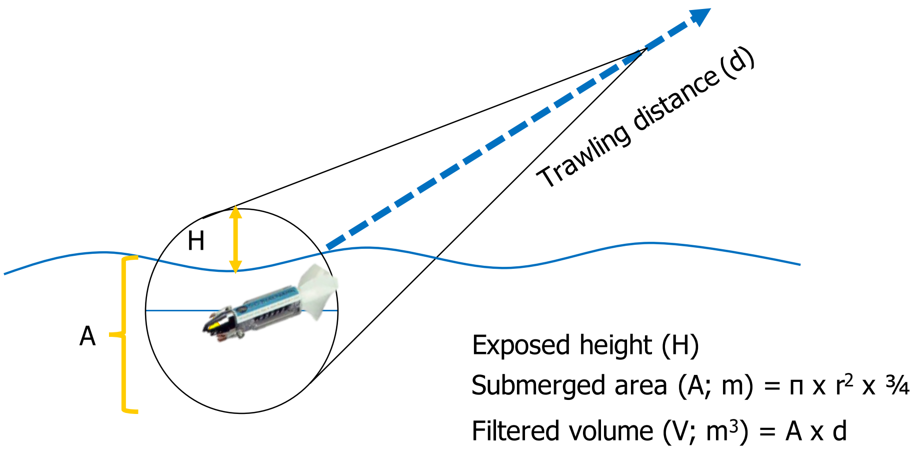

Figure 1. Correct position of the plankton net during sampling.

In the event of not having a flowmeter, the volume calculation will be done as follows:

a) Locate a homogeneous section of the riverbed, preferably straight, with a single channel and without obstructions such as submerged logs or large rocks, avoiding the presence of pools, deep holes, waterfalls, or cascades. Ideally, it should be a site with a uniform flow.

b) Take two references in the section, upstream and downstream, where the reading will be taken, measuring the actual distance of the section through a previous test.

c) Use a stopwatch and preferably a floating object (e.g., an orange). Release the object gently upstream from the first reference point, in the center of the channel, and record the time it takes to travel the total distance.

4. Rinse the net from the outside inward (Fig. 2) to concentrate the sample in the collector. Avoid letting water enter the mouth of the net.
5. Collect the water sample in a plastic/glass container previously rinsed with Milli Q water (or distilled water).
6. Label the sample: (SITErep1_ddmmyyyy). Keep refrigerated in a refrigerator for short-term analysis.
7. Repeat sampling (2 replicates, SITErep2_ddmmyyyy). It is mandatory to wash the net carefully between replicates. To facilitate cleaning, immerse the net several times in the body of water with the valve open, then rinse with water from the outside of the net as much as possible. Close the collector valve again after finishing.
8. Take note of any sources of microplastic contamination. Example: worn boat paint, plastic material used during sampling, clothing (color included).

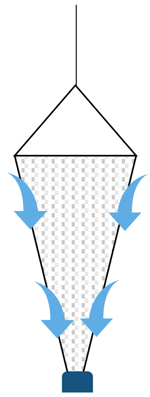

Figure 2. Example of how to clean the net with water to concentrate the sample.

Table 1. Estimated height of the plankton net that should remain above the water surface based on the diameter of the net used during the collection of a microplastics sample (3/4 of the net area is submerged).

Net Diameter (cm)	Plankton Net Surface (cm^2)	Net Area Surface Out of Water (cm^2)	% surface out of water	Height (H) of the net out of water (cm)
30	706,9	178,4	25,23	**9**
35	962,1	242,8	25,23	**10,5**
40	1256,6	317,1	25,23	**12**
45	1590,4	401,3	25,23	**13,5**

Calculate the **required filtering time** according to your net parameters as follows:

Time (s) = [Volume (m^3)] / [Area (m^2) x 3/4 x Velocity (m/s)]

Table 2. Filtration times for a 20 m3 sample in lotic environments according to the diameter of the plankton net (considering 3/4 of the net submerged) and the flow velocity at the net position.

Speed (cm s^{-1})	Plankton Net Diameter (cm) 30		Plankton Net Diameter (cm) 40	
	Filtering time (decimals)	**Filtering time (min; sec)**	**Filtering time (decimals)**	**Filtering time (min; sec)**
30	21,0	**21; 0**	11,8	**11; 48**
40	15,7	**15; 42**	8,8	**8; 48**
50	12,6	**12; 36**	7,1	**7; 06**
60	10,5	**10; 30**	5,9	**5; 54**
70	9,0	**9; 0**	5,1	**5; 06**
80	7,9	**7; 54**	4,4	**4; 24**
90	7,0	**7; 0**	3,9	**3; 54**
100	6,3	**6; 18**	3,5	**3; 30**

CAUTION: If the environment exhibits flow velocities around 1 m/s and is turbid, there is a possibility of the net becoming clogged and breaking easily. Under such circumstances, it is recommended to perform filtration for approximately 1 minute, rinse the net, collect the sample, and repeat this process.

B- Sites with low flow (< 0.5 m depth): BUCKET AND PLANKTON NET

1. To optimize the selection of the sampling point, whenever feasible, prioritize reaching the deepest accessible location by walking. If the bottom material has been resuspended, wait before proceeding with the sample collection process.
2. Using a metal bucket, take a sample of surface water at a depth of 20 cm in the deepest accessible area (Fig. 3). Before collection, calibrate the bucket to an appropriate volume for sampling (for example, a volume of 8 liters will require at least 25 collections). Submerge the bucket carefully to avoid resuspending materials from the bed, margins, or any objects in the channel that may retain materials, such as organic matter.
3. Carefully pour the water into the plankton net, ensuring that it does not spill outside the mouth of the net (Fig. 3).
4. Repeat this process as many times as necessary until filtering at least 200 liters of water. If you believe you can filter a larger volume, it is recommended to do so.
5. Rinse the net from the outside inward (Fig. 2) to concentrate the sample in the collector. Avoid letting water enter the mouth of the net.
6. Collect the water sample in a plastic/glass container previously rinsed with Milli-Q water (or distilled water).
7. Label the sample: (SITErep1_ddmmyyyy). Keep refrigerated for short-term analysis.
8. Repeat sampling (2 replicates, SITErep2_ddmmyyyy). It is mandatory to wash the net carefully between replicates. To facilitate cleaning, immerse the net several times in the body of water with the valve open, then rinse with 5 liters of distilled water. Close the collector valve again after finishing.
9. Take note of any sources of microplastic contamination. Example: worn boat paint, plastic material used during sampling, clothing (color)

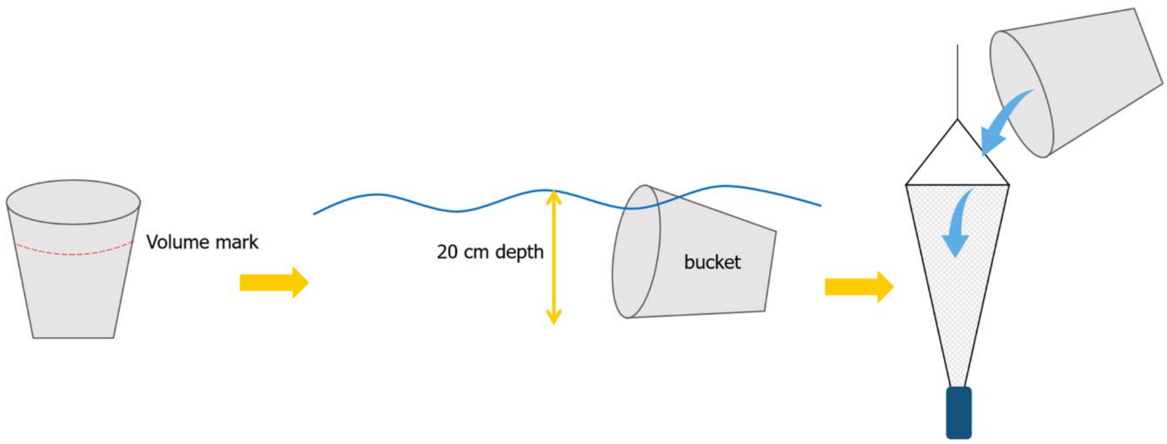

Figure 3. Procedure to obtain a water sample with a bucket.

4.4. Metadata

- Geographical coordinates
- Altitude above sea level (m.a.s.l.)
- Depth (Maximum and/or average)
- Water body area
- Waterfront development
- Basin area
- Length of residence
- Monthly rainfall
- Average effluent of the river flow
- Percentage of land uses in the basin
- Population density in the basin
- Distance to the nearest city center
- Distance to roads, recreation centers, etc.
- Presence of sewage or discharges
- Main activities (tourism, fishing, etc.)
- Physicochemical parameters: Chlorophyll-a (Chl-a) concentration, salinity, turbidity, Secchi depth, organic matter concentration (in situ measurements or at least average data from the literature)

5. Other links of interest:

Reference Guides:
GESAMP, 2019
http://www.gesamp.org/publications/guidelines-for-the-monitoring-and-assessment-of-plastic-litter-in-the-ocean
Michida et al., 2020 (version 1.2)
https://www.env.go.jp/content/000170474.pdf